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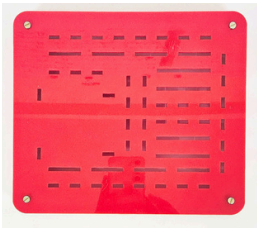
I'm a ROBO SCIENTIST

SESSION NAME

Session

9

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1


2


3


4


5

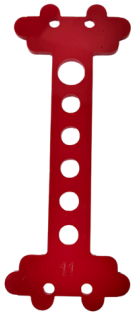

6


7


8


9


10


11


1


Spacers

2


L Clamp

3

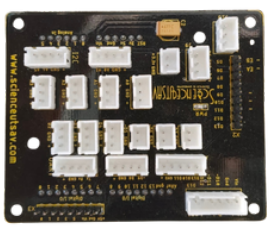

Spacer Screws

4


Nuts

5


Screw

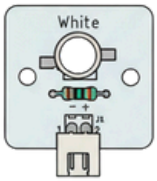
6


Arduino With Shield

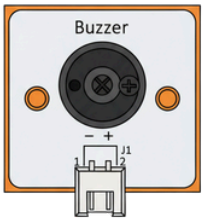
7


Arduino cable

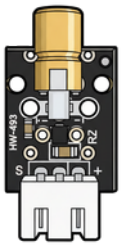
ELECTRONIC COMPONENTS



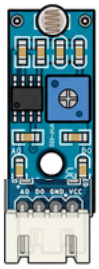
1-watt LED
A 1-watt LED lamp is an energy-saving light that uses less electricity. It gives bright light while producing very little heat. It is commonly used in small lamps, indicators, and decorative lighting.



BUZZER PCB
A buzzer is an electronic component that generates sound when electric current flows through it. It is commonly utilized in alarms, alerts, and notification systems to provide auditory signals.



Laser diode
A Laser diode is a semiconductor device that produces a focused beam of light. It works by converting electrical energy into laser light. Laser diodes are used in barcode scanners, fiber-optic communication, and CD/DVD players.



LDR sensor
An LDR sensor is a sensor that detects the intensity of light. Its resistance changes according to the amount of light falling on it. It is used in automatic street lights, alarms, and light-controlled devices.



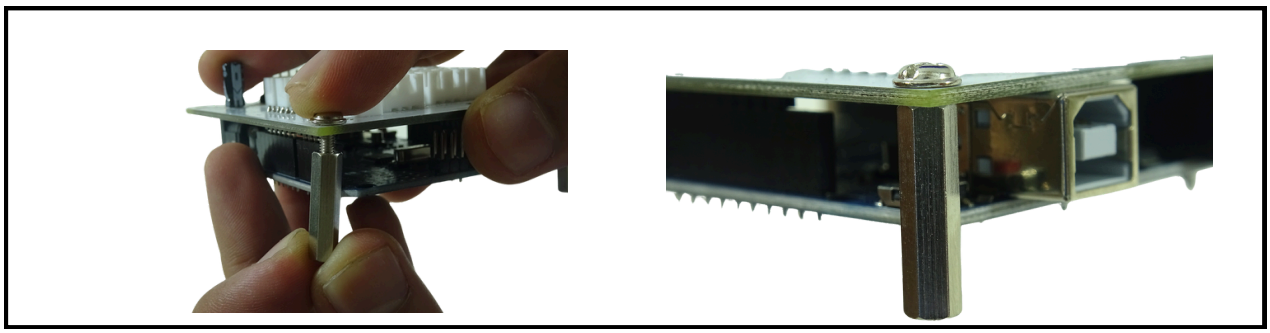
Push Switch
A push switch is a button-style switch that operates only when pressed.
It provides temporary input in circuits, such as starting a device or triggering an action.



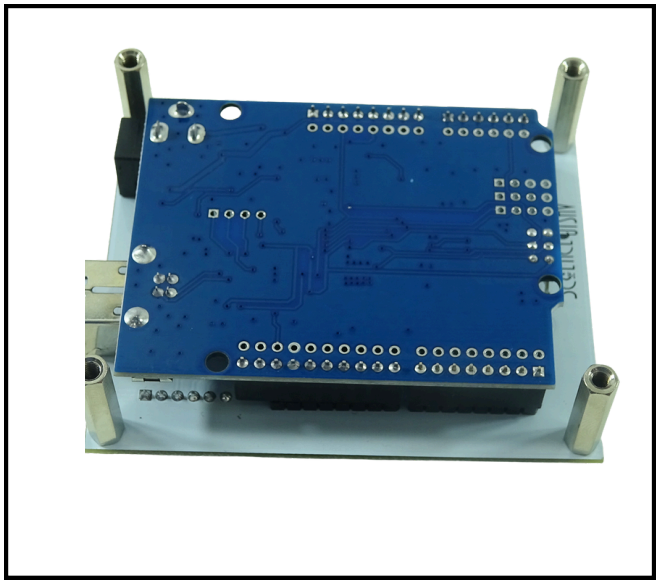
7-segment display
7 seg display electronic display device used to show numbers. It has seven LED segments arranged in the shape of the digit 8. By turning ON and OFF different segments, it can display numbers from 0 to 9. It is commonly used in digital clocks, calculators, counters, meters, and electronic device

Procedure :

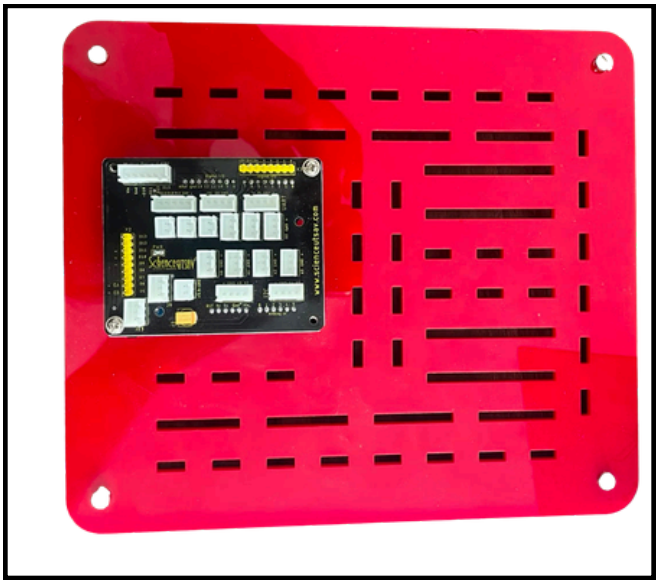
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw



Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



Step 3 : Place the Arduino Shield to the base board as shown below

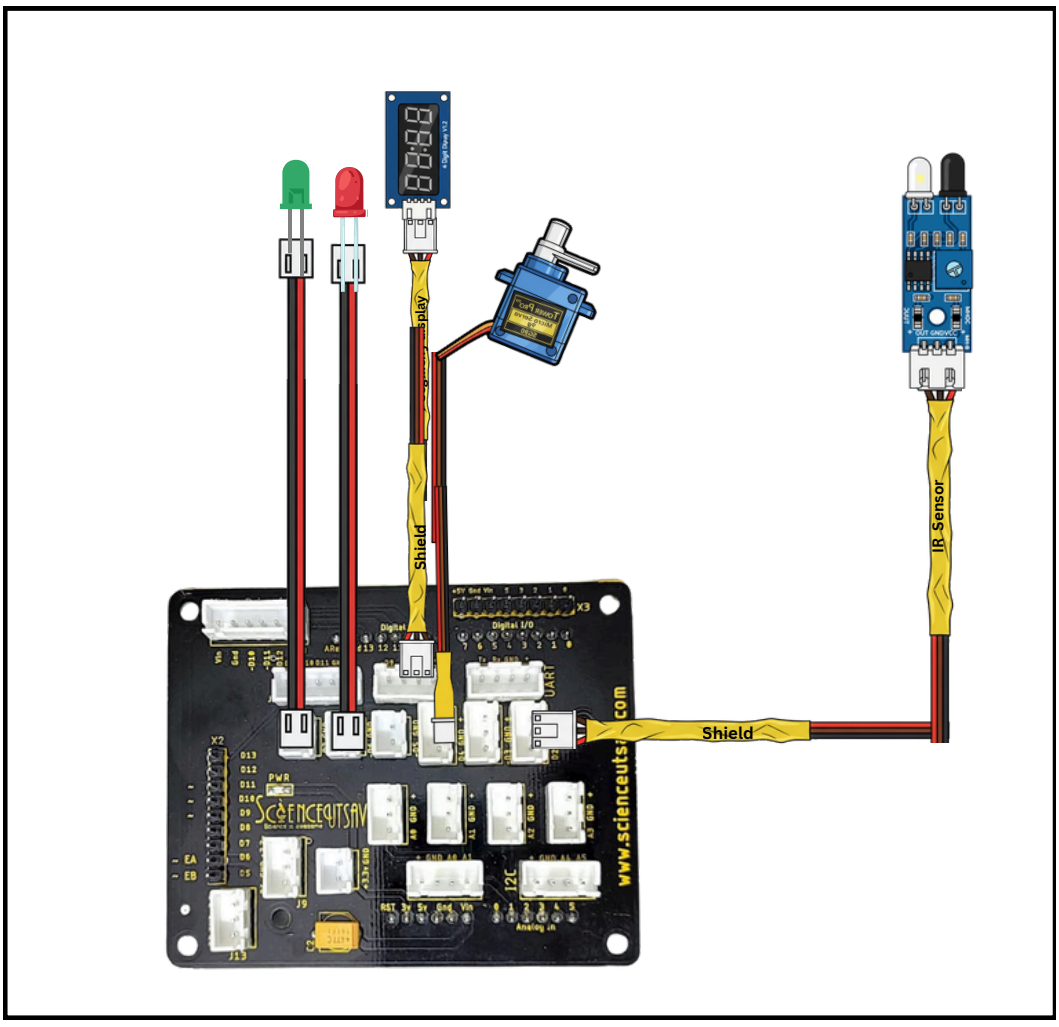


Step 4 : Fix it with the spacer screw from the bottom as shown below



BEGINNER PROJECT

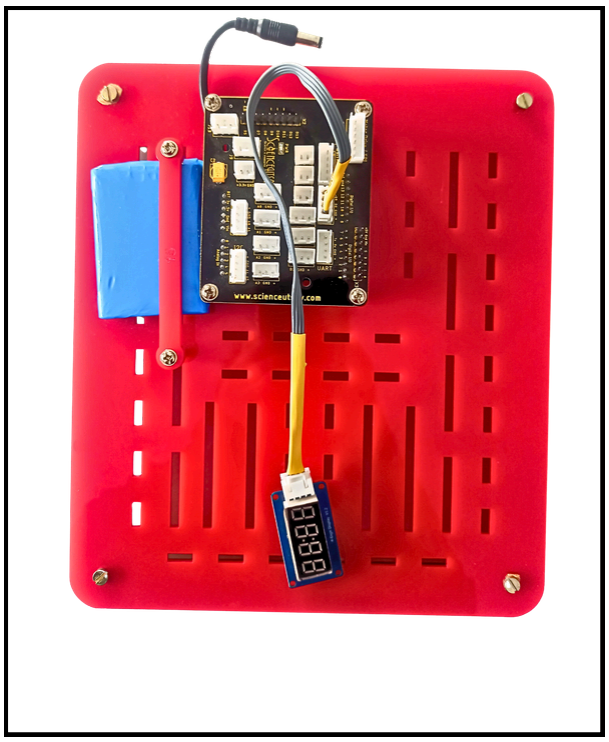
The circuit diagram of the project is given below



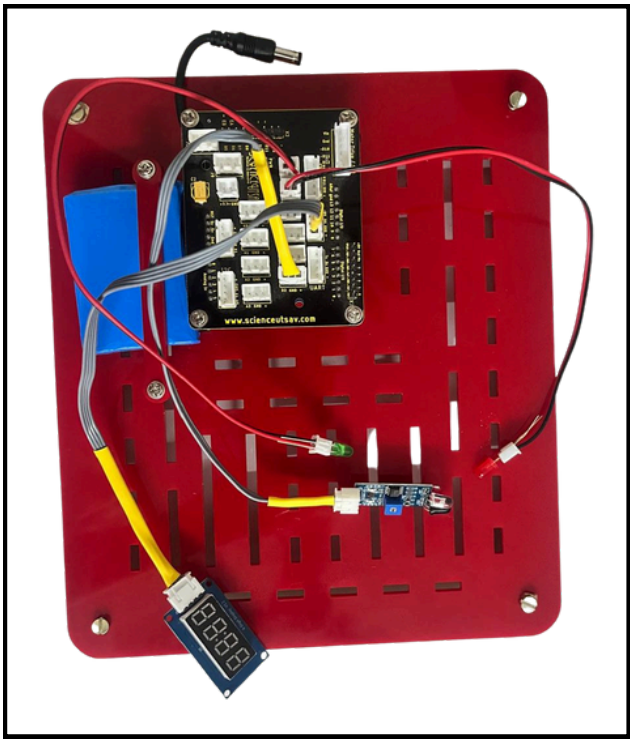
CONNECTIONS:

IR Sensor - D2
Servo Motor - D4
7 Segment Display - D8, D9
Red LED - D6
Green LED - D7

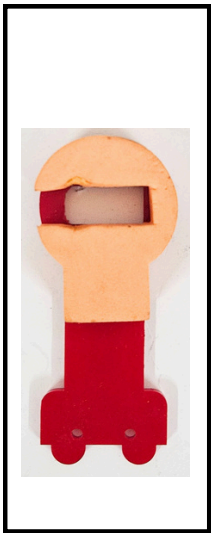
Step 1- Attach the 7 segment display to the aurdino as shown



Step 2- Attach the led lights and 7 segment display as shown



Step 3- Attach the template to a servo motor stand, place the servo motor on the stand, attach the leds on the template as well







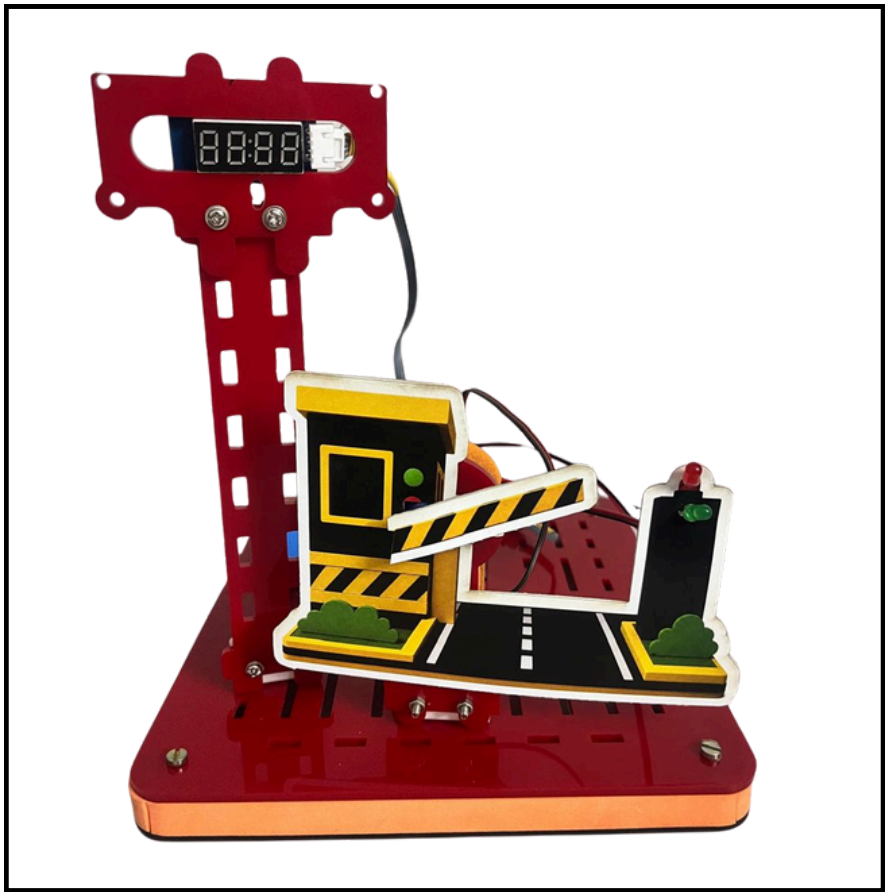
Step 4- Fix the boom gate template to the servo motor



Step 5- Fix the display stand to the base



Step 6- Fix the template to the board and upload the code



Code

```
// Session 9 Grade 6
// Smart Gate
// Vehicle detected = count increases, gate opens, green LED ON
// No vehicle = gate closed, red LED ON

#include <Servo.h>
#include <TM1637Display.h>

const int irSensor = 2;
const int servoPin = 4;
const int redLED = 10;
const int greenLED = 11;
const int CLK = 9;
const int DIO = 8;

const int IR_DETECTED = LOW;  // change to HIGH if opposite

Servo gateServo;
TM1637Display display(CLK, DIO);

int vehicleCount = 0;
bool lastIR = HIGH;

void setup() {
pinMode(irSensor, INPUT);
pinMode(redLED, OUTPUT);
pinMode(greenLED, OUTPUT);

gateServo.attach(servoPin);
gateServo.write(0);

display.setBrightness(7);
display.showNumberDec(0, true);
```

```
digitalWrite(redLED, HIGH);
digitalWrite(greenLED, LOW);
}

void loop() {
bool currentIR = digitalRead(irSensor);

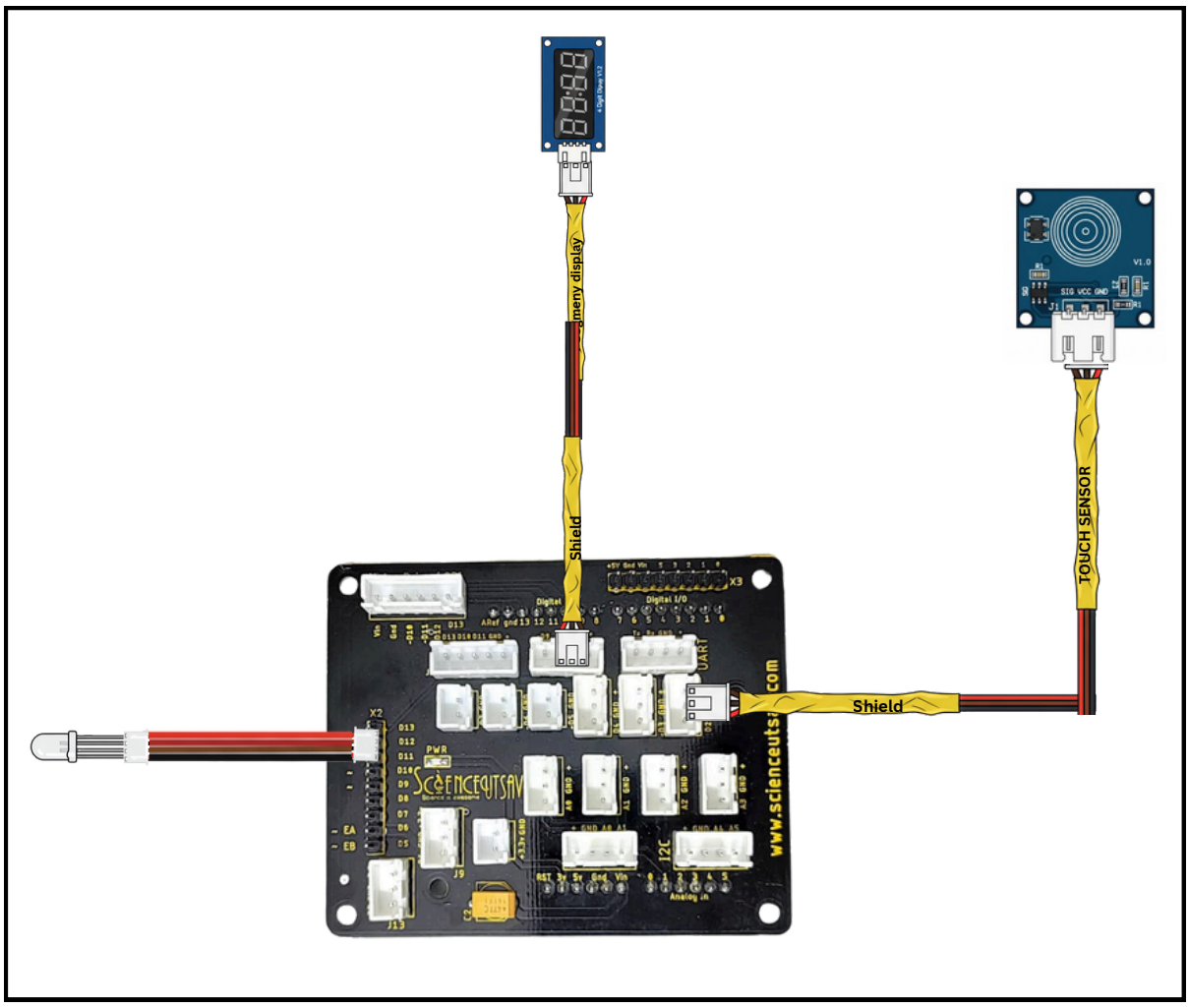
if (currentIR == IR_DETECTED && lastIR != IR_DETECTED) {
vehicleCount++;
if (vehicleCount > 9999) vehicleCount = 0;
display.showNumberDec(vehicleCount, true);

digitalWrite(redLED, LOW);
digitalWrite(greenLED, HIGH);
gateServo.write(90);
delay(3000);
gateServo.write(0);
digitalWrite(greenLED, LOW);
digitalWrite(redLED, HIGH);
delay(300);
}

lastIR = currentIR;
}
```

INTERMEDIATE PROJECT

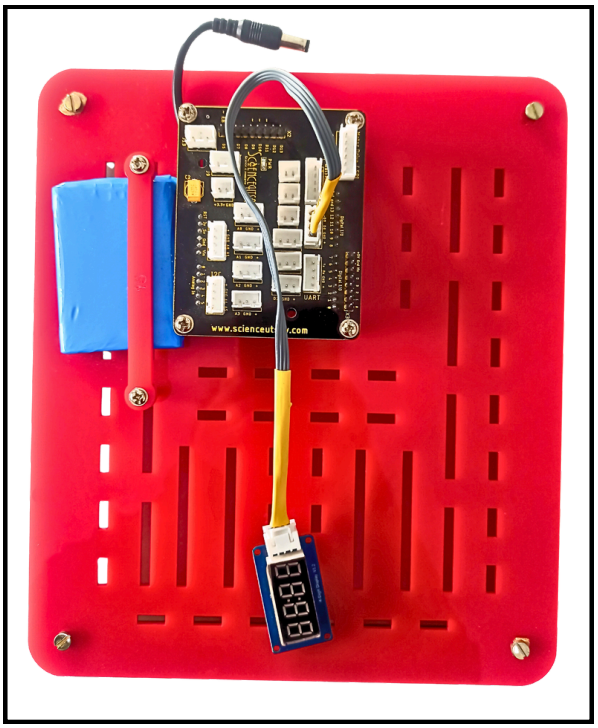
The circuit diagram of the project is given below



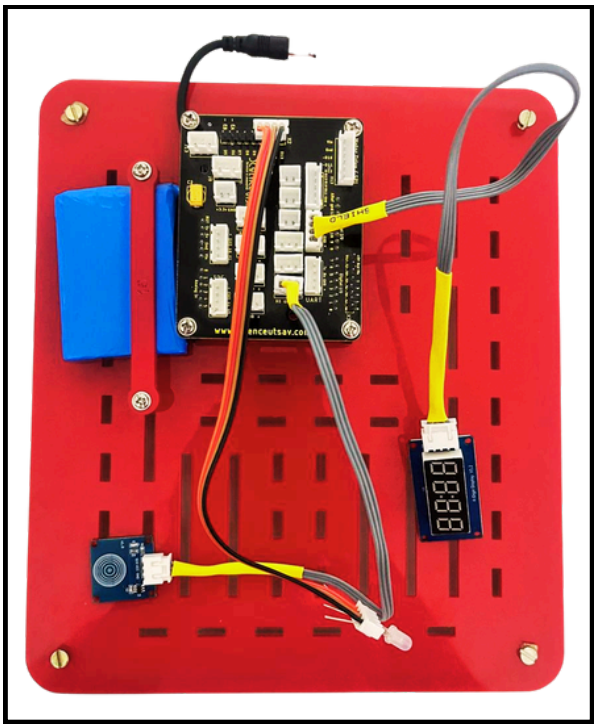
CONNECTION:

1. Touch sensor - D2
2. 7-segment display - D8, D9
3. RGB led - D13, D12, D11, D10

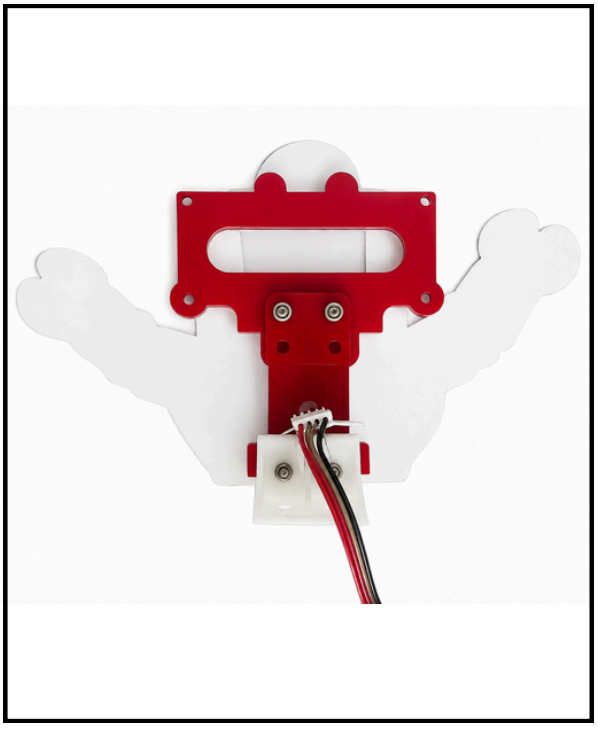
Step 1- Attach the 7 segment display to the aurdino as shown



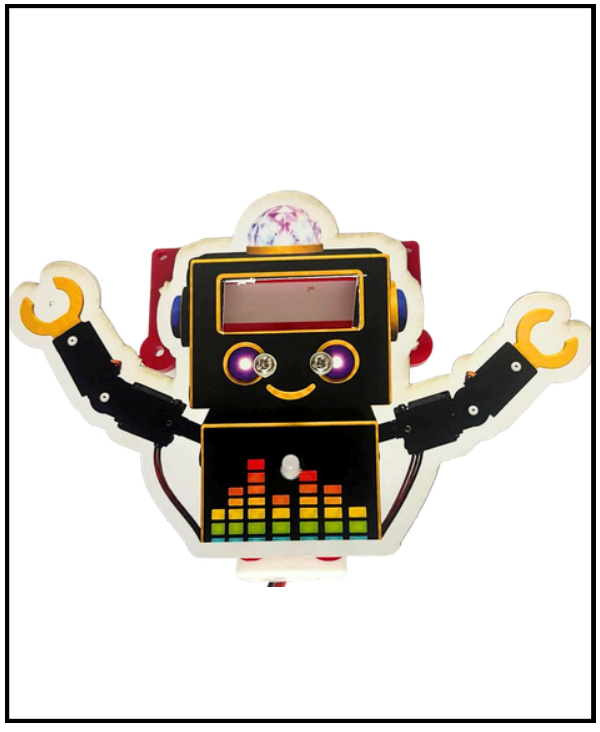
Step 2- Attach the touch sensor and RGB LED to the aurdino as well



Step 3- Attach the stand to the template as shown



Step 4- Fix the RGB LED to the top of the template



Step 5- Fix the template to the board and attach the display to it



Code

```
// Session 9 Grade 7
 // Dancing Robo
 // Touch sensor touched = servo dances, LED blinks, display pattern changes

 #include <Servo.h>
 #include <TM1637Display.h>

 const int touchSensor = 2;
 const int servoPin = 4;
 const int led = 6;
 const int CLK = 9;
 const int DIO = 8;

 const int TOUCH_ACTIVE = HIGH; // change to LOW if opposite

 Servo roboServo;
 TM1637Display display(CLK, DIO);

 void setup() {
 pinMode(touchSensor, INPUT);
 pinMode(led, OUTPUT);

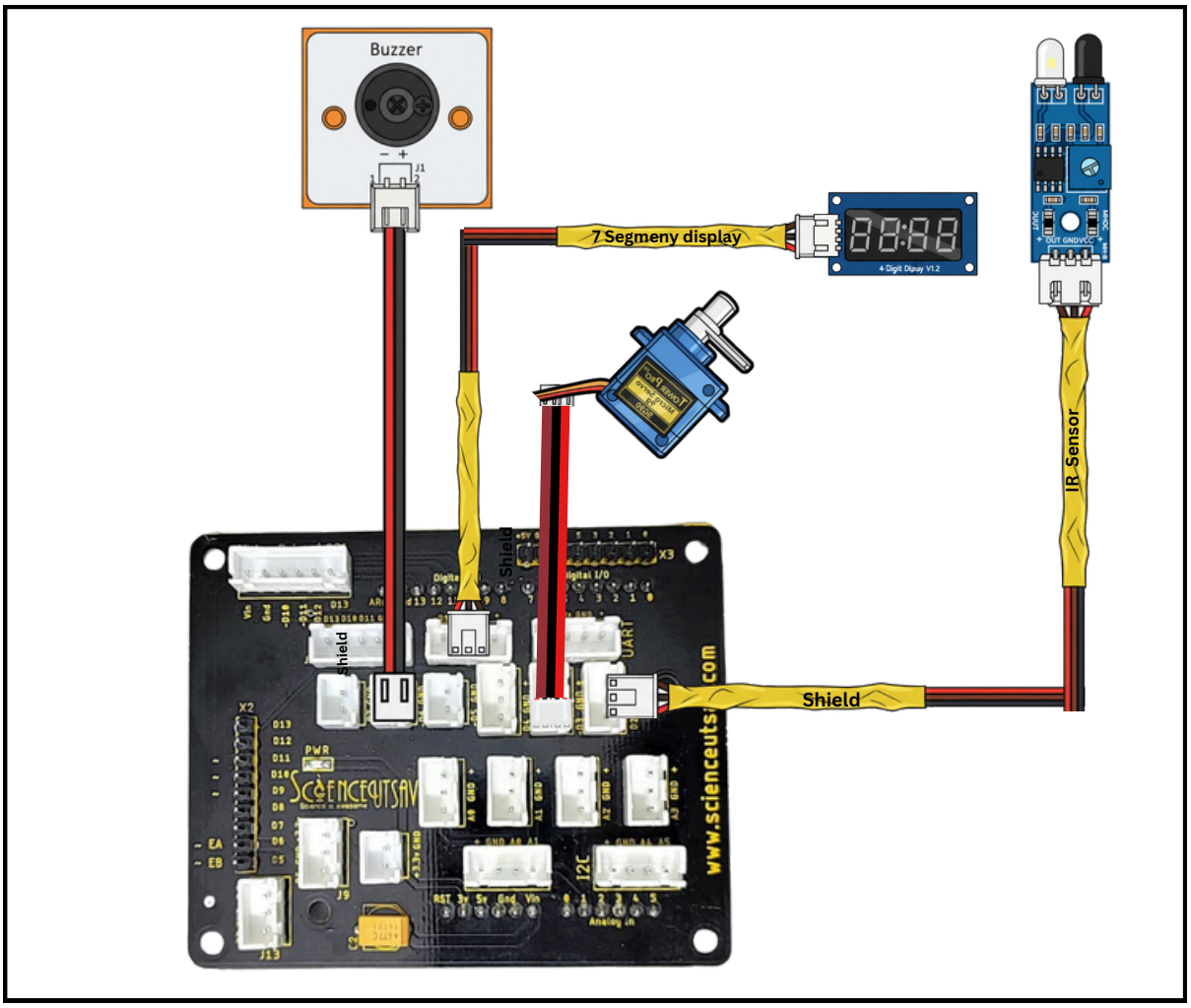
 roboServo.attach(servoPin);
 roboServo.write(90);

 display.setBrightness(7);
 display.showNumberDec(0, true);
 }

 void loop() {
 if (digitalRead(touchSensor) == TOUCH_ACTIVE) {
  for (int i = 1; i <= 5; i++) {
    display.showNumberDec(i, true);
    digitalWrite(led, HIGH);
    roboServo.write(40);
    delay(300);
    digitalWrite(led, LOW);
    roboServo.write(140);
    delay(300);
  }
  roboServo.write(90);
  display.showNumberDec(0, true);
 }
 }
```

ADVANCE PROJECT

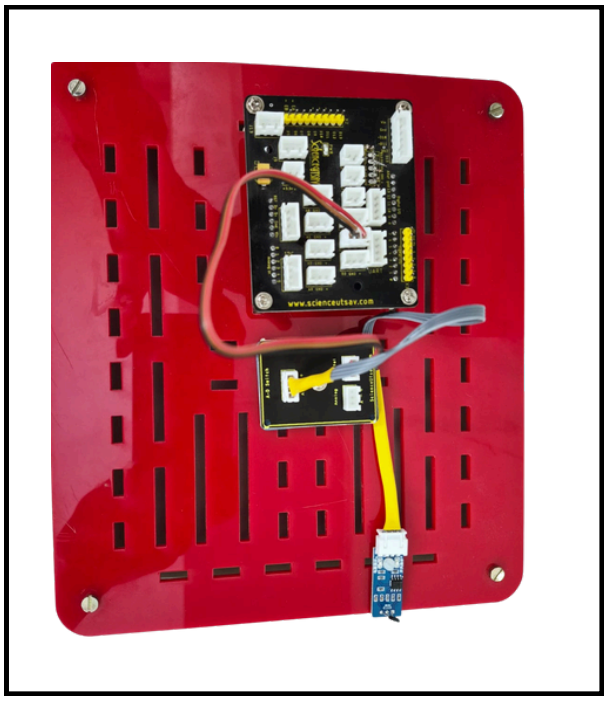
The circuit diagram of the project is given below



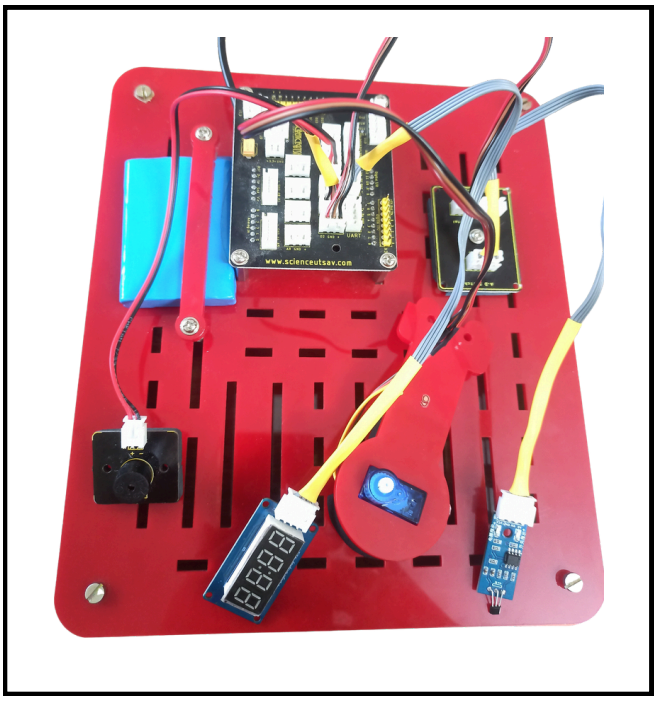
CONNECTION:

1. Servomotor- D3
2. Buzzer- D6
3. 7-segment dislay- D8, D9
4. IR sensor- D2

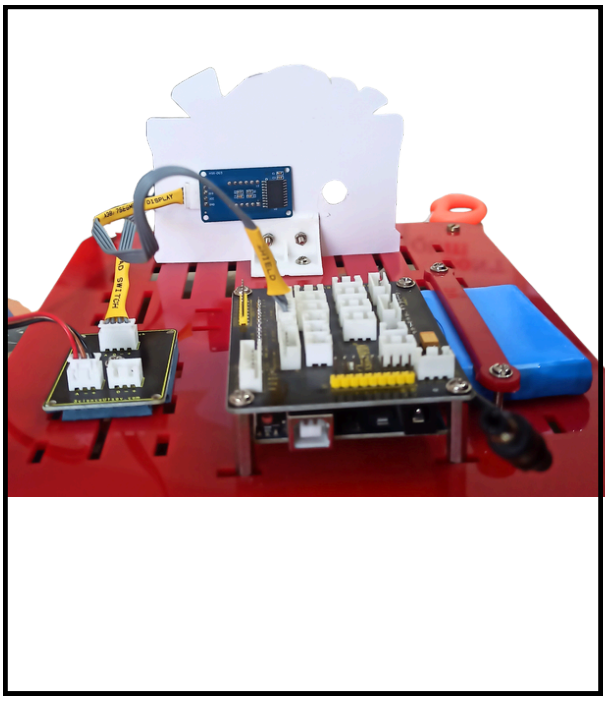
Step 1- Connect the AD switch to the shield and attach the magnetic sensor to it



Step 2- Fix the display, buzzer and servomotor as shown



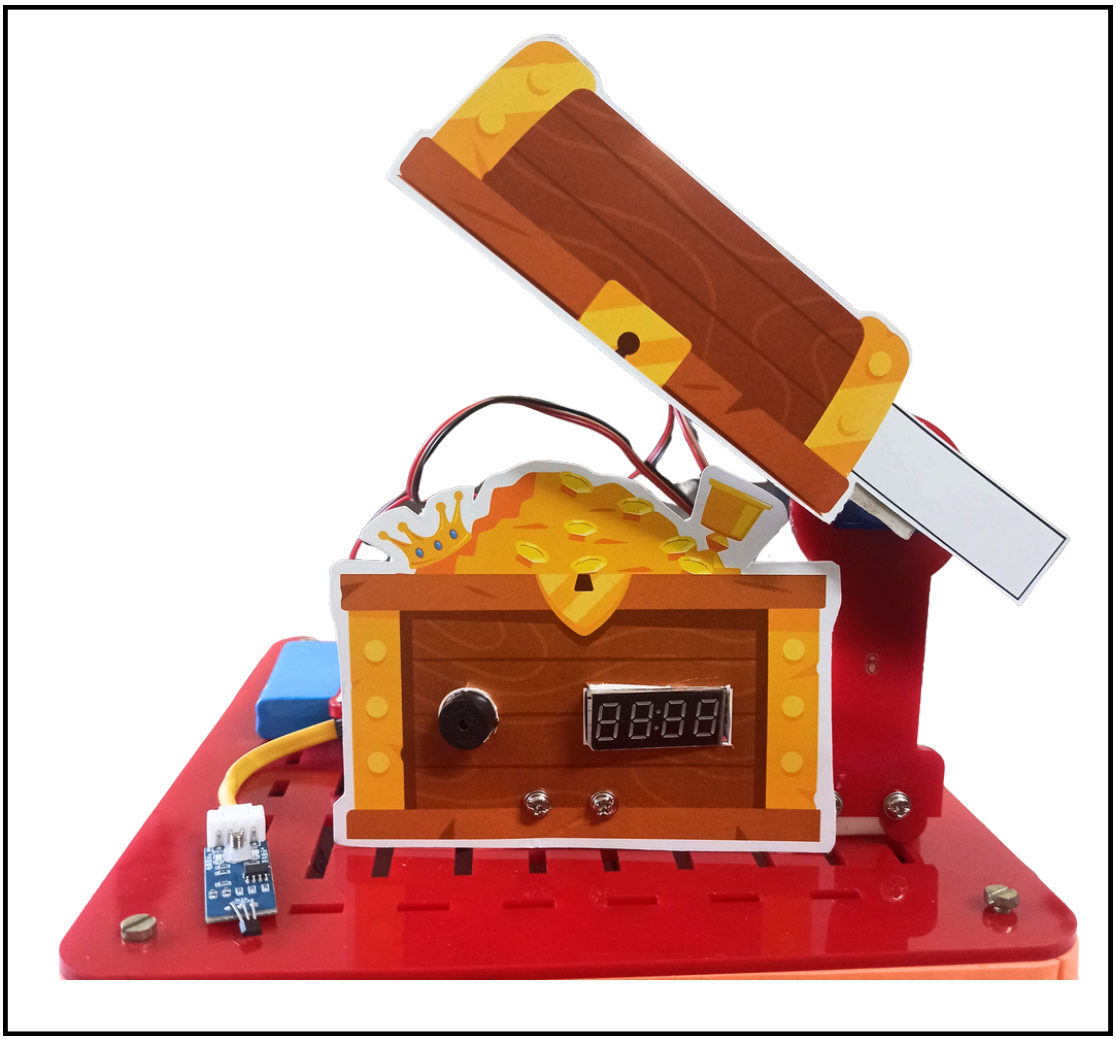
Step 3- Fix the template using L-clamp and attach the display to it



Step 4- Attach the buzzer and the servomotor stand as shown



Step 5- Fix the other template to the servomotor



Code

```
// Session 9 Grade 8
 // Treasury Lock Counter
 // Magnet near Hall sensor = treasury unlocks and count increases
 // Push button resets count
 // Hall sensor is used as DIGITAL sensor with digitalRead(A0)

 #include <TM1637Display.h>

 const int hallSensor = A0;
 const int pushButton = 5;
 const int led = 6;
 const int CLK = 9;
 const int DIO = 8;

 const int MAGNET_DETECTED = LOW; // change to HIGH if opposite

 TM1637Display display(CLK, DIO);

 int openCount = 0;
 bool lastMagnet = HIGH;

 void setup() {
 pinMode(hallSensor, INPUT);
 pinMode(pushButton, INPUT_PULLUP);
 pinMode(led, OUTPUT);

 display.setBrightness(7);
 display.showNumberDec(0, true);
 }

 void loop() {
 bool magnetNow = digitalRead(hallSensor);
```

```
if (magnetNow == MAGNET_DETECTED) {
 digitalWrite(led, HIGH);
 } else {
 digitalWrite(led, LOW);
 }

 if (magnetNow == MAGNET_DETECTED && lastMagnet !=
MAGNET_DETECTED) {
 openCount++;
 if (openCount > 9999) openCount = 0;
 display.showNumberDec(openCount, true);
 delay(300);
 }

 if (digitalRead(pushButton) == LOW) {
 openCount = 0;
 display.showNumberDec(0, true);
 delay(300);
 }

 lastMagnet = magnetNow;
 }
```